

Computer Architecture

CSE 511/ECE 511

Mid-Semester Examination

DATE: 14 October, 2020

Instructions:

- The duration of the exam is **60 minutes**.
 - The exam is for a total of **50 Marks**.
 - **All questions** are compulsory.
 - Please list down your assumptions (if any) clearly.
 - Please make sure to properly differentiate different stages while writing pipeline diagrams.
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Q1. Consider a processor with 5 pipeline stages. The delay of each pipeline stage are 4 ns, 5 ns, 10 ns, 10 ns and 7ns respectively. The registers residing between the pipeline stages has a delay of 2 ns. Determine the approximate speedup of the pipelined implementation as compared to its non-pipelined implementation? [5 Marks]

Q2. Consider the information given in Table 1 regarding Machine A and Machine B. A program takes 50,000 instructions when run on Machine A. How many instructions would be needed to run the same program on Machine B, so that both the machines have same execution time for the same program. [5 Marks]

	CPI	Clock rate
Machine A	1.5	600MHz
Machine B	2.5	800MHz

Q3. Consider a 32-bit machine with set-associative cache. The cache size is 16KB and it is a 4-way set-associative cache. The length of memory location is 1 byte and block size is 16 Bytes.

- Calculate the total number of sets present in the cache.
- Draw the 32-bit memory address showing the tag, index and offset bits.
- Determine the set number where the content of memory address 10FFCFF_(HEX) is mapped. Give explanation or steps. [(2 + 4 + 4) Marks]

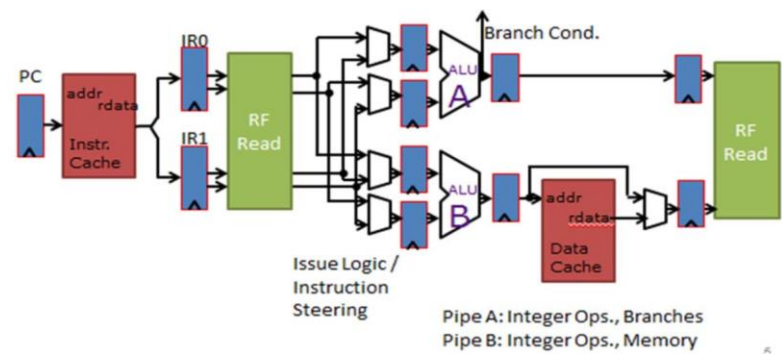
Q4. Consider a fully bypassed 5-stage RISC-V processor with the following pipeline stages: **F, D, X, M, W**. Whenever a branch instruction is encountered, it is resolved in the **X** Stage of the pipeline. The **F** Stage always speculates that the next PC is PC+4. Here, concentrate on only conditional branches and ignore any unconditional jumps. Consider the following set of instructions given below. Note that value of register R0 is 0.

0x1000: ADDI R4, R0, 9
0x1004: LD R5, 0(R0)
0x1008: ADD R5, R5, R4
0x100c: BEQ R4, R5, 0x1000
0x1010: LD R7, 4(R6)
0x1014: ADD R5, R7, R5
0x1018: SUB R1, R2, R3

Draw the pipeline timing diagram (resource usage diagram will not be considered) of the above instructions. You can draw the pipeline till seven instructions. Calculate how many clock cycles will be required to complete till seven instructions. [15 Marks]

Q5. Draw pipeline diagram (resource usage diagram will not be considered) of the following code running on the in-order two-way superscalar processor in the figure. Each pipeline stage takes 1 cycle. Assume that branches execute in pipeline A, loads/stores in pipeline B. Assume full bypassing. Calculate how many clock cycles will be required to complete all the instructions. [15 Marks]

Inst1: ADD R14, R16, R17
 Inst2: SUB R1, R2, R3
 Inst3: SUB R2, R3, R5
 Inst4: LD R11, 0(R2)
 Inst5: ADDI R12, R13, 1
 Inst6: LD R15, 4(R2)
 Inst7: LD R15, 4(R15)
 Inst8: SUB R2, R9, R11
 Inst9: ADD R3, R11, R12
 Inst10: ADD R4, R6, R7



[Figure of Two-Wide In-order Superscalar Processor Pipeline]